

Appendix G: Cross-Platform Universality of $\bar{C}^*$ — A Negative Result

UIP Phase 1 Review Packet · Falsified preregistered claim · The "Unified" designation is retired · Companion to preprint DOI: [10.5281/zenodo.21246247](https://doi.org/10.5281/zenodo.21246247) · July 2026
Code: `appendix_g_constant_transfer.py` at github.com/derekhone/uip-phase1-testbeds

Headline result: FALSIFIED. The claim that the constant $\bar{C}^* = 1/1.45$ and the crossover law $s^*(\gamma)$ are platform-independent ("unified") was tested by constructing a structure-matched classical analog of the quantum testbed. The preregistered prediction was classical crossovers within ± 0.05 of $s^*(0.15) = 0.2628$ and $s^*(0.25) = 0.3420$. **No crossover exists in the classical system at any tested coupling** — always-correct dominates everywhere. The quantity that $\bar{C}^*$ balances (measurement back-action cost of correction) is not merely different classically; it is *undefined*. The universality claim, and with it the "U" in UIP, is retired. Cross-domain applications (biological, cosmological, energetic) proposed in early versions of the framework are unsupported and positively contraindicated by this result.

1. Claim under test

Appendices D/E confirmed the zero-free-parameter law $s^*(\gamma) = \arcsin(\sqrt{(\gamma(1-\bar{C}^*)/\bar{C}^*)})$, $\bar{C}^* = 1/1.45$, in a quantum feedback testbed. The framework's founding ambition — a *unified* account of coherence across physical domains — requires that this balance not be an artifact of quantum mechanics. The sharpest available test: rebuild the testbed classically, structure-matched element for element, and check whether the same constant locates a crossover.

2. The structure-matched classical analog

Quantum element	Classical counterpart
Qubits A (relay), B (shielded target)	Two noisy oscillators A, B
ℓ_1 coherence $C(\rho)$	Signal amplitude / phase-order parameter
Partial SWAP, $\tau = \sin^2(s)$	Beat-exchange coupling with identical $\tau = \sin^2(s)$
Dephasing $e^{-\gamma}$, alternating attack	Phase diffusion decay $e^{-\gamma}$, same schedule
Partial re-preparation of A ($f = 0.5$)	Partial reference-signal re-injection into A ($f = 0.5$)
Noisy estimates, 30 seeds	Identical sensing noise and seeding

Preregistered (before running):

P-G1: the classical system exhibits a correction crossover within ± 0.05 of $s^*(0.15) = 0.2628$.

P-G2: and within ± 0.05 of $s^*(0.25) = 0.3420$.

Kill-condition: either point outside its window — or no crossover at all — falsifies universality of $\bar{C}^*$.

3. Result

γ	Predicted classical crossover	Measured	Verdict
0.15	0.2628 ± 0.05	none exists — correction advantage positive at all s	FAIL
0.25	0.3420 ± 0.05	none exists — correction advantage positive at all s	FAIL

P-G1, P-G2: FAIL. Always-correct dominates never-correct at every tested coupling strength in the classical system. There is no sign change, hence no crossover, hence nothing for $\bar{C}^*$ to locate.

4. Diagnosis: why the constant cannot transfer

The quantum crossover exists because correction is *costly* there: re-preparing the relay is a measurement-like operation that disturbs the off-diagonal phase structure the relay is carrying toward the target (back-action). The crossover is the point where delivered benefit equals that disturbance cost — $\bar{C}^* = 1/1.45$ is the coherence level at which they balance.

Classical re-injection of a reference signal has **no such cost**: adding clean signal to a classical oscillator does not disturb what it is already carrying. With the cost term identically zero, $G > L$ always, and correction is unconditionally beneficial. The balance $\bar{C}^*$ encodes is a specifically quantum trade-off. One could rescue "transfer" by inserting a hand-built disturbance penalty into the classical model — but that would import the quantum feature under test by fiat and would not constitute independent cross-platform validation.

5. Consequences for the framework

- **The "Unified" claim is retired.** The framework's confirmed content (Appendices A, C, D, E) is quantum-feedback-specific. The honest name for what survives is a *quantum-feedback boundary-predicting bookkeeping method*.
- **Cross-domain extensions are contraindicated.** Applications to biological coherence, cosmology, or energy systems assumed the platform-independence tested here; this record now weighs against them, not merely fails to support them.
- **The confirmations are sharpened, not weakened.** Knowing *why* $\bar{C}^*$ exists (back-action balance) makes the $s^*(\gamma)$ law more physically intelligible, and makes hardware validation (where back-action is real and measurable) the natural next test.

6. Honest scope

The classical analog is one structure-matched instantiation among many possible; reviewers proposing alternative classical systems with a *native* (non-inserted) correction cost are invited to test them — a classical crossover at $\bar{C}^* = 1/1.45$ in such a system would reopen the question and would be reported as such. All simulations are by the framework's author; code is public and seeded. This negative result is published unedited at its preregistered threshold.

"A false balance is an abomination to the LORD, but a just weight is His delight." — Proverbs 11:1